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/*
 * C program to Implement Stack using Array
 */

#include <stdio.h>
#define MAX_SIZE 20

// function prototyping
void push(int);
int pop(void);
int peek(void);
int is_empty();
int is_full();
void display();

struct stack
{
    int stk[MAX_SIZE];
    int top;
};

struct stack st;

int main()
{
    st.top = -1;

    char userContinue = 'Y';
    int userChoice;
    while (userContinue == 'Y' || userContinue == 'y') {
        printf("\n----- STACK DATA STRUCTURE ----- \n\n");
        printf("\n1. Push the data");
        printf("\n2. Pop the data");
        printf("\n3. Peek the data");
        printf("\n4. Display Stack");
        printf("\n5. Exit");

        printf("\n\nEnter Your Choice: ");
        scanf("%d", &userChoice);

        switch (userChoice)

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{
    case 1:
        printf("\n\nEnter Data: ");
        scanf("%d", &userChoice);
        push(userChoice);
        break;
    case 2:
        if (!is_empty())
        {
            int data = pop();
            printf("\nValue %d was popped", data);
        }
        else
        {
            printf("\n\tStack is Empty!");
        }
        break;
    case 3:
        if (!is_empty())
        {
            int data = peek();
            printf("\nValue %d was peeked", data);
        }
        else
        {
            printf("\n\tStack is Empty!");
        }
        break;
    case 4:
        display();
        break;
    case 5:
        printf("\n\tProgram Terminated!\n");
        return 1;
        break;
    default:
        printf("\n\tInvalid Choice!");
}

printf("\n\nDo you want to continue? ");
fflush(stdin);
scanf(" %c", &userContinue);
}

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        return 0;
    }

int is_full()
{
    if (st.top == MAX_SIZE)
    {
        return 1;
    }

    return 0;
}

int is_empty()
{
    if (st.top == -1)
    {
        return 1;
    }
    return 0;
}

void push(int data)
{
    if (is_full())
    {
        printf("\nStack Overflow!");
    }
    else
    {
        st.stk[++st.top] = data;
        printf("\n* %d was inserted", data);
    }
}

int pop()
{
    return st.stk[st.top--];
}

int peek()
{
    return st.stk[st.top];
}

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}

void display()
{
    if (is_empty())
    {
        printf("\n\tStack is Empty!");
        return;
    }
    for (int i = st.top; i >= 0; i--)
    {
        printf("\n\t%d", st.stk[i]);
    }
}
```